# Dilan Karaguler

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## Research Interests

Applied algebraic topology and topological data analysis; path homology and Mayer (N-nilpotent) homology of directed graphs and path complexes; persistent homology and persistent Laplacians; topology-informed graph machine learning and geometric deep learning, with applications to molecular, dynamical and engineering data.

## Education

**Michigan State University** — East Lansing, MI, US
Ph.D. in Mathematics (Candidate) — August 2022 – May 2027 (Expected)
- Advisors: Prof. Ekaterina Rapinchuk and Prof. Guowei Wei

**Middle East Technical University (METU)** — Ankara, Türkiye
M.S. in Mathematics — September 2019 – September 2021
- Thesis: *A Survey on Multidimensional Persistence Theory*
- Advisor: Assoc. Prof. Semra Pamuk

**Middle East Technical University (METU)** — Ankara, Türkiye
B.S. in Mathematics, Honor Student — June 2014 – June 2019

## Publications

**D. Karaguler** and G.-W. Wei. *Mayer Path Homology.* Foundations of Data Science, accepted (2026). doi:10.3934/fods.2026025, arXiv:2605.16525.

**D. Karaguler** and G.-W. Wei. *Persistent Mayer Path Homology and Laplacian.* Preprint (2026), to appear on arXiv.

**D. Karaguler** and G.-W. Wei. *Directed Topology for Protein–Nucleic Acid Flexibility.* In preparation (2026).

**D. Karaguler.** *A Survey on Multidimensional Persistence Theory.* M.S. Thesis, Middle East Technical University (2021).

## Research Experience

**Graduate Researcher** — August 2022 – Present
Michigan State University — East Lansing, MI, US
- Develop Mayer path homology, a homology theory for directed graphs built from N-nilpotent differentials on path complexes; published in *Foundations of Data Science* (with G.-W. Wei).
- Extended it to persistent Mayer path homology and the Mayer path Laplacian, with stability theorems and a molecular application (preprint, 2026).
- Implement topological feature extraction (path complexes, Mayer homology and Laplacians, persistent homology) in Python with GUDHI, NumPy and PyTorch, and combine these features with graph neural networks.

**Research Mentor** — May 2023 – June 2023
Topology RTG, Michigan State University — East Lansing, MI, US
- Mentored an undergraduate researcher on a project in Morse theory: project design, methodology and research presentation; the student went on to major in mathematics.

## Industry Experience

**Software Engineer Intern** — May 2026 – August 2026
Intel Corporation — Oregon, US
- Built an end-to-end graph transformation pipeline in Python that extracts schematic topology directly from chip layout files, without pre-existing schematic metadata.
- Modeled low-level device instances as graph nodes and extracted a local subgraph around each instance to capture its connectivity.
- Developed graph aggregation and embedding routines producing a vector representation per device, used to classify components as analog, digital or fill devices.
- Worked with Design Yield and Manufacturability engineers to evaluate the prototype for automated structural recognition in layout verification.

**Graduate Technical Intern** May 2025 – August 2025
Intel Corporation Oregon, US

- Performed spatial pattern analysis on semiconductor chip layer images with KLayout and Python to identify and cluster recurring structures across layout datasets.
- Applied graph neural networks to search for and classify similar patterns across multi-layer designs.
- Partnered with cross-functional yield teams to bring ML-based pattern recognition into automated yield-analysis workflows.

## Teaching Leadership & Mentoring

**Lead Teaching Assistant** August 2024 – Present
Department of Mathematics, Michigan State University East Lansing, MI, US

- **2026 – 2027:** Co-lead, with Eloy Moreno-Nadales, a yearlong workshop series for first-year graduate TAs: introductions, *Three Aspects of Teaching*, mock teaching with peer feedback, and a graduate student panel.
- **Fall 2025 (MTH 132):** Prepared recitation worksheets, gathered TA feedback for instructors, and substituted for recitation leaders.
- **Spring 2025 (MTH 133):** Ran TA meetings, prepared weekly worksheets, and set up Teams tools to track grading deadlines.
- **Fall 2024:** Ran ULA orientation and mock-teaching sessions, and observed ULAs and GTAs to give feedback.

**Graduate Lead, Center for Instructional Mentoring (CIM)** Summer 2024
Department of Mathematics, Michigan State University East Lansing, MI, US

- Mentored first-time graduate instructors through weekly check-ins and non-evaluative feedback.
- Created a teaching handbook from interviews with senior GTAs on each course's expectations.

**Mentor, Academic Culture and English Skills (ACES) Program** Summer 2024
Michigan State University East Lansing, MI, US

- Mentored incoming international graduate TAs on U.S. classroom culture and cross-cultural communication; organized social events.

**Supervisor, Math Learning Center (MLC)** Fall 2022 – Spring 2023, Fall 2024
Michigan State University (Teaching Exemption Fellowship, 2022 – 2023) East Lansing, MI, US

- Managed how tutors were distributed among students and arranged the room to match demand.

**Student feedback:** dilankaraguler.github.io/teaching/what-students-say

## Courses Taught

**MTH 314 — Matrix Algebra with Computational Applications** Fall 2026
Instructor (1 section), Michigan State University East Lansing, MI, US

- Prepare and deliver lectures, facilitate group work, and hold help-room hours.
- Recorded per-question quiz scores to send each student personalized feedback on topics to review and problems to practice, and showed coordinators how quiz versions affected grades.

**MTH 299 — Transitions** Fall 2025
Teaching Assistant (2 sections), Michigan State University East Lansing, MI, US

- Co-taught with Ester Sgallova: weekly recitations, homework solutions, grading and help-room hours.

**MTH 314 — Matrix Algebra with Computational Applications** Spring 2025
Teaching Assistant (1 section), Michigan State University East Lansing, MI, US

- Graded homework and the final exam; ran quiz and exam reviews and help-room hours.

**MTH 133 — Calculus II** Fall 2024
Teaching Assistant (1 section) and pod leader, Michigan State University East Lansing, MI, US

- Led a recitation; as pod leader, ran recitation-leader meetings and wrote the weekly worksheets.

**MTH 235 — Differential Equations** Spring 2024
Teaching Assistant (3 sections), Michigan State University East Lansing, MI, US

- Wrote recitation notes with summaries and practice problems; graded quizzes and held help-room hours.

**MTH 133 — Calculus II** Fall 2023
Teaching Assistant (3 sections: 2 online, 1 in person), Michigan State University East Lansing, MI, US

- Prepared weekly lecture summaries; graded and proctored quizzes; held help-room hours and followed up with online students.

**MTH 133 — Calculus II** Summer 2023
Teaching Assistant (1 online asynchronous section), Michigan State University East Lansing, MI, US

- Contacted students weekly by email and tracked their course activity.

**MTH 154, 153, 120, 119 — Calculus I and II** January 2020 – June 2022
Graduate Teaching Assistant, Middle East Technical University Ankara, Türkiye
- MTH 154 (Spring 2022) and MTH 153 (Fall 2021) for Mathematics Students; MTH 120 (Spring 2021) and MTH 119 (Fall 2020) for Engineering Students.
- Led recitations and office hours; graded and proctored exams.

**MTH 154 — Calculus II for Mathematics Students** Spring 2019
Undergraduate Teaching Assistant (homework grader), Middle East Technical University Ankara, Türkiye
- Graded homework for more than 300 students.

**MTH 153 — Calculus I for Mathematics Students** Fall 2018
Undergraduate Teaching Assistant (homework grader), Middle East Technical University Ankara, Türkiye
- Graded homework for more than 300 students.

**MTH 116 — Introduction to Abstract Algebra** Spring 2017
Undergraduate Teaching Assistant (recitation instructor), Middle East Technical University Ankara, Türkiye
- Conducted recitations for 180 students with my own recitation materials.

## Curriculum Development

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| MTH 314 Matrix Algebra (Instructor): lectures and group-work activities | Fall 2026 |
| MTH 132 Calculus I (Lead TA): weekly group activities with Desmos explorations | Fall 2025 |
| MTH 299 Transitions: homework solutions as a recitation leader | Fall 2025 |
| MTH 133 Calculus II (pod leader, Lead TA): weekly recitation worksheets for all sections | Fall 2024 – Spring 2025 |
| Teaching handbook for first-time graduate instructors, Center for Instructional Mentoring | Summer 2024 |
| MTH 235 Differential Equations: recitation notes with summaries and practice problems | Spring 2024 |
| MTH 133 Calculus II: weekly lecture summaries for recall sessions | Fall 2023 |
| METU MTH 154 Calculus II: recitation questions and homework assignments | Spring 2022 |
| METU MTH 153 Calculus I: recitation questions and homework assignments | Fall 2021 |
| METU MTH 116 Introduction to Abstract Algebra: recitation materials | Spring 2017 |

## Teaching Development & Educational Research

**Program Evaluator & Instructional Researcher** May 2026
AI for Advanced Semiconductor Manufacturing Workshop, Michigan State University East Lansing, MI, US
- Designed pre/post surveys and observation protocols, and analyzed feedback with quantitative and thematic coding.
- Wrote an evaluation report with curriculum recommendations grounded in adult learning theory.

**MTH 879 Teaching College Mathematics (graduate course)** Fall 2024
Michigan State University East Lansing, MI, US

**Colleges Online Learning Academy (COLA) Scholar and workshops** Summer 2024
Michigan State University Portfolio
- Completed workshops on online course design; built a teaching portfolio website and an online course plan.

**Certification in College Teaching Institute (CCTI), two-day workshop** May 2024
Michigan State University East Lansing, MI, US

## Talks

**Integrating AI to Lecture Prep (upcoming)** November 2026
Mathematics Teaching Seminar, *Sparks: Teaching Lightning Talks by Graduate Instructors* MSU

**Persistent Mayer Path Homology and Laplacian (upcoming)** November 2026
Applied Mathematics Student Seminar, Michigan State University East Lansing, MI, US

**Mayer Path Homology** April 2026
TDA Seminar, Dept. of Computational Mathematics, Science and Engineering, MSU East Lansing, MI, US

**Interaction Topology in TDA** October 2025
Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU East Lansing, MI, US

**Mayer Topology in TDA** October 2025
Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU East Lansing, MI, US

**From Data to Barcodes** April 2021
Graduate Student Seminar, Middle East Technical University Ankara, Türkiye

## Poster Presentations

***Mayer Path Topology and Persistence*** — September 2025
SIAM Great Lakes Section Meeting, Illinois Institute of Technology — Chicago, IL, US

***Mayer Path Topology and Persistence*** — March 2025
Mathematics of Data, Dynamics, and Life Sciences Conference, UC Irvine — Irvine, CA, US

## Projects

**Persistent Mayer Homology with Dynamic Mode Decomposition** — April 2026
Python, NumPy, SciPy, GUDHI — GitHub
- Built a reproducible pipeline combining Vietoris–Rips filtrations, Mayer homology and Dynamic Mode Decomposition (DMD) to track time-varying topological features.
- Computed Mayer Betti trajectories on synthetic 3D point dynamics; a rank-10 DMD model reconstructed a 500-frame dataset with about 3.6% relative error and supported multi-step forecasting.

**Homology-Fed Neural Network Framework** — November 2024
Python, PyTorch, GUDHI, NumPy — GitHub
- Wrote a Python library computing Mayer homology and path-complex Laplacians on graph-structured data.
- Used the resulting topological features for molecular property prediction on the QM9 dataset with graph neural networks.

**Cellular Automata on Curved Surfaces** — March 2024
Python, NumPy, Matplotlib — GitHub
- Simulated Conway's Game of Life on a square, torus and sphere with surface-specific neighbor rules, and analyzed how board size and initial density affect population persistence.

## Awards, Fellowships & Scholarships

Doctoral Student Summer Internship Fellowship, The Graduate School, Michigan State University — Summer 2026
**Excellence in Teaching Award**, Department of Mathematics, Michigan State University — May 2025
Colleges Online Learning Academy (COLA) Fellowship, Michigan State University — Summer 2024
Teaching Exemption Fellowship, Michigan State University — 2022 – 2023
RONESANS Scholarship, RONESANS — 2018 – 2019
UNIGENC Scholarship, UNIGENC Association — 2015 – 2019
Merit Scholarship, Turkish Ministry of National Education — 2010 – 2014

## Service, Leadership & Outreach

**Graduate Representative, NatSci Student Advisory Council (SAC)** — 2025 – 2026
College of Natural Science, Michigan State University — East Lansing, MI, US

**Graduate Student Representative, Graduate Studies Committee** — 2024 – 2025
Department of Mathematics, Michigan State University — East Lansing, MI, US
- Represented mathematics graduate students on the committee and co-planned Responsible Conduct of Research (RCR) workshops.
- Hosted town halls, met prospective students, and directed students with concerns to department resources.

**Member, First-Generation/Low-Income (FGLI) Group** — September 2024 – May 2025
Department of Mathematics, Michigan State University — East Lansing, MI, US
- Took part in community gatherings and social events with first-generation and low-income undergraduates.

**Organizing Committee Member** — April 2024
Graduate Student Topology and Geometry Conference, Michigan State University — East Lansing, MI, US
- Handled speaker communication, scheduling and room assignments; moderated sessions.

**President / Board Member** — August 2022 – August 2025
Association for Women in Mathematics (AWM), MSU Chapter — East Lansing, MI, US
- Organized seminars, career workshops, academic-life sessions and social events for women in mathematics.

**President** — February 2015 – December 2018
METU Math Club — Ankara, Türkiye
- Organized seminars, workshops and reading groups; prepared first-year students to lead abstract algebra and number theory discussions.

### Outreach & Volunteering

Proctor, Math Kangaroo Team Competition, Department of Mathematics, MSU (upcoming) — November 2026
Volunteer, Math Night, Marble Elementary School, East Lansing — November 2026
Volunteer, MSU Student Food Bank, Michigan State University — Spring 2024

## Conferences & Workshops Attended

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| Agentic AI (hot topic course), AIDMM NSF Research Traineeship, Michigan State University | Fall 2026 |
| AI, Data and PDE Workshop, Michigan State University | April 2025 |
| Graduate Student Topology and Geometry Conference, Michigan State University | April 2024 |
| Mid-Atlantic Topology Conference, Northeastern University | March 2024 |
| SIAM Great Lakes Section Meeting, Michigan State University | October 2023 |
| Computational Persistence, Purdue University | September 2023 |
| Randomness in Topology and its Applications, IMSI, Chicago | March 2023 |
| LMS Workshop: Applied Algebraic Topology, Queen Mary University of London (online) | February 2022 |
| Topological Insights in Neuroscience, IMSI (online) | May 2021 |
| Topological Data Analysis, IMSI (online) | April 2021 |

## Technical Skills

**Programming**: Python, R, SQL
**Methods**: Graph Neural Networks, Graph Transformers, Topological Data Analysis, Persistent Homology, Vietoris–Rips Filtrations, Dynamic Mode Decomposition, Supervised/Unsupervised Learning, Model Evaluation
**Libraries**: PyTorch, PyTorch Geometric, NetworkX, GUDHI, NumPy, SciPy, Pandas, Scikit-Learn, TensorFlow, Keras, Matplotlib, XGBoost, LightGBM
**Tools**: Git, Linux/Unix, DuckDB, SQLite, Jupyter, KLayout, LaTeX
**Domain Knowledge**: Design for Manufacturability (DFM), Semiconductor Yield Analysis, Schematic Extraction, Chip Layout Analysis

## References

Available upon request.